

Efficient gene delivery admitted by small metabolites specifically targeting astrocytes in the mouse brain

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The development of efficient and targeted methods for delivering DNA *in vivo* has long been a major focus of research. In this study, we introduce a gene delivery approach admitted by small metabolites (gDAM) for the efficient and targeted delivery of naked DNA into astrocytes in the adult brains of mice. gDAM uses a straightforward combination of DNA and small metabolites, including glycine, L-proline, L-serine, L-histidine, D-alanine, Gly-Gly, and Gly-Gly-Gly, to achieve astrocyte-specific delivery of naked DNA, resulting in transient and robust gene expression in these cells. Using gDAM, we successfully co-deliver the PiggyBac transposon and the CRISPR-Cas9 system to induce long-term overexpression of the oncogene EGFRvIII and knockout of tumor suppressor genes *Nf1*, *Pten*, and *Trp53* in astrocytes, leading to the development of astrocyte-derived gliomas in immunocompetent mice. Furthermore, gDAM facilitates the delivery of naked DNA to peripheral glioma astrocytes. The overexpression of interferon- β and granulocyte-macrophage colony-stimulating factor in these peripheral glioma astrocytes significantly prolongs the overall survival of mice bearing 73C glioma cells. This approach offers a new perspective on developing gene delivery systems that specifically target astrocytes to meet the varied needs of both research and gene therapy. The innovative strategy behind gDAM is expected to provide fresh inspiration in the quest for DNA delivery to other tissues, such as skeletal muscle and skin.

INTRODUCTION

The intricate nature of the brain is defined by a myriad of cell types, each characterized by distinctive functionalities and morphologies. Astrocytes, which constitute approximately 20%–40% of cells in the CNS, play a pivotal role in the development and maintenance of the CNS. They contribute to the integrity of the blood-brain barrier, provide metabolic support to neurons, and facilitate neurogenesis and synaptogenesis.^{1,2} The activation and dysregulation of astrocytes in neurodegenerative diseases are increasingly recognized as key mech-

anisms linked to the progression of neuronal damage.^{3–5} Moreover, astrocytes have been shown to play an important role in the brain tumor microenvironment,^{6,7} implicating them in the progression and proliferation of brain tumors.

Gene delivery techniques are critical for exploring the manifold functions of astrocytes, and viral vectors are favored vehicles for gene delivery due to their high transfection efficiency.^{8,9} Adeno-associated virus (AAV) vectors are considered the safest option among viral vectors; however, their potential to trigger undesired immune responses still poses significant challenges and concerns in clinical trials.^{10,11} Additionally, AAV suffers from a notoriously limited packaging capacity due to its small capsid size (approximately 260 Å in diameter). As a result, only genomes up to approximately 4.7 kb can be efficiently packaged and produced, which limits the inclusion of large cell-type-specific promoters and large transgenes.¹² Another critical issue associated with viral vectors is the lack of cell-specific targeting. Despite the development of various astrocyte-specific promoters, including GfaABC1D, glial fibrillary acidic protein (GFAP), and Aldh1l1, options for precise genetic manipulation of astrocytes are currently limited.^{13–15} A recently reported toolbox of AAVs has dramatically increased astrocytic specificity through multiple miRNA-targeting sequences designed to degrade transgenes in neurons and endothelial cells.¹⁶ However, the addition of miRNA-targeting sequences may severely compromise the already limited packaging capacity of AAV vectors. Therefore, there is a pressing need to develop highly accurate and effective technologies for the genetic manipulation of astrocytes to advance the understanding of their intrinsic functions

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and astrocyte-mediated multicellular interactions, as well as to pioneer novel therapeutic approaches for CNS disorders.

To address the limitations of viral vectors in brain-targeted gene therapy and therapeutic genome editing, nonviral vectors have gained increasing attention as promising alternatives. Several nonviral methods have been developed, including chemical-based nanoparticles and naked DNA.^{17–19} Non-viral vectors are considered to have significant potential to overcome challenges associated with viral vectors, such as production difficulties, packaging size, and concerns about immunogenicity.^{20,21} However, the suboptimal transfection efficiency of nanoparticles in the brain severely limits their applicability in gene therapies for CNS diseases. While the tropism of nonviral vectors can be enhanced through continuous optimization of the programmable properties of nanomaterials,²² highly cell-type-specific and safe nonviral vectors remain extremely rare in the CNS.

The use of naked DNA as a gene delivery vector has been a part of genetic research and biotechnology for several decades. Interest in its potential for gene delivery began to increase significantly in the early 1990s, after demonstrations that plasmid DNA could be delivered directly into muscle tissue and subsequently express the intended genes.^{23–26} Over the years, various strategies, including electroporation and hydrodynamic pressure have been developed to enhance the cellular uptake efficiency of naked DNA vectors.^{27–31} Despite the simplicity of naked DNA as a gene delivery system, its effectiveness has been significantly hampered by low delivery efficiency and well-known cytotoxicity when applied to the adult brain. Therefore, there is an urgent need to further enhance the efficiency and specificity of nonviral vectors for brain-targeted delivery.

Herein, we introduce a novel gene delivery method called gene delivery approach admitted by small metabolites (gDAM), which effectively delivers naked DNA to achieve transient and robust gene expression exclusively in astrocytes within the murine brain. gDAM is notable for its ability to carry large genetic payloads and deliver multiple genes simultaneously. This method is both practical and effective, representing a significant advancement over traditional AAV-based techniques and other nonviral delivery systems. gDAM shows great potential for enhancing basic research on astrocytes and for developing therapeutic strategies for CNS diseases.

RESULTS

Glycine-mediated delivery of naked DNA targets specifically in astrocytes

The quest to deliver genetic material with precision and efficiency to targeted brain cell populations is pivotal for advancing our understanding of neurological functions and developing therapeutic interventions. To address this challenge, we identified glycine, a simple metabolic molecule, as an effective facilitator for delivering naked DNA plasmids to brain cells. Our approach involved administering a mixture of a plasmid vector encoding the fluorescent protein mScarlet and 300 mM glycine to the midbrain of mice (Fig-

ure 1A). Five days after administration, we observed a striking aggregation of fluorescently transfected cells near the injection sites, indicating successful gene uptake and expression in the brain cells (Figure 1B).

To evaluate the indispensable role of glycine in the gDAM approach, we varied its concentration from 0 to 300 mM. The results demonstrated a positive correlation between glycine concentration and fluorescent protein expression. In contrast, no expression was observed when the plasmid DNA was mixed with PBS lacking glycine (Figure 1C). Subsequent quantification revealed that the number of tdTomato-expressing cells increased with the glycine concentration, ranging from approximately 17 ± 4 transfected cells at 25 mM to $5,269 \pm 517$ cells at 300 mM (Figures 1D and 1E). Additional experiments using glycine with a plasmid encoding eGFP under the cytomegalovirus (CMV) promoter showed a similar trend of concentration-dependent increases in fluorescence intensity (Figure S1).

The glycine transporter type-1 (GlyT1), found in presynaptic neurons and astrocytes, is considered the primary regulator of extracellular glycine levels in the brain.^{32,33} Blocking GlyT1 can slow the clearance of extracellular glycine, prolonging its presence in the interstitial space and maintaining its concentration. To ascertain the pivotal role of glycine, we added bitopertin, a GlyT1 inhibitor, to the gDAM mixture containing 100 mM glycine. The addition of bitopertin increased the number of transfected cells and the fluorescence intensity of eGFP by 2.9 and 3.8 times, respectively (Figure S2), reinforcing the importance of glycine availability in the extracellular milieu for successful gDAM gene delivery.

To confirm the cell types transfected by gDAM, confocal images captured at higher magnification revealed that all transfected cells displayed characteristic astrocyte morphology (Figure 1F). To further verify that the gDAM-transfected cells are astrocytes, Aldh1l1-eGFP mice, which express eGFP specifically in brain astrocytes, were crossbred with Ai9 transgenic mice. In the presence of Cre recombinase, which is particularly sensitive to off-target expression, Cre-mediated recombination results in the expression of tdTomato in the crossbred mice. The co-expression of eGFP and tdTomato indicates that the transfected tdTomato-labeled cells can be identified as astrocytes (Figure S3A). Subsequently, a mixture of plasmid vectors expressing Cre recombinase and 300 mM glycine was injected into the midbrain of the crossbred mice. The confocal images revealed that all gDAM-transfected tdTomato-labeled cells were also eGFP positive, confirming that the gDAM-transfected cells were indeed astrocytes (Figures 1G and S3B–S3C). In comparison, when an AAV virus encoding Cre recombinase controlled by the astrocyte-specific promoter (GfaABC1D) was injected into the crossbred mice, only 62% of the AAV-transfected cells were eGFP positive (Figures 1G and 1H). These findings underscore that gDAM specifically targets astrocytes, achieving 100% specificity (Figure 1H). To further corroborate these observations, immunofluorescence staining for microglial (ionized calcium-binding adaptor molecule 1 [Iba1]) (Figure S4A),

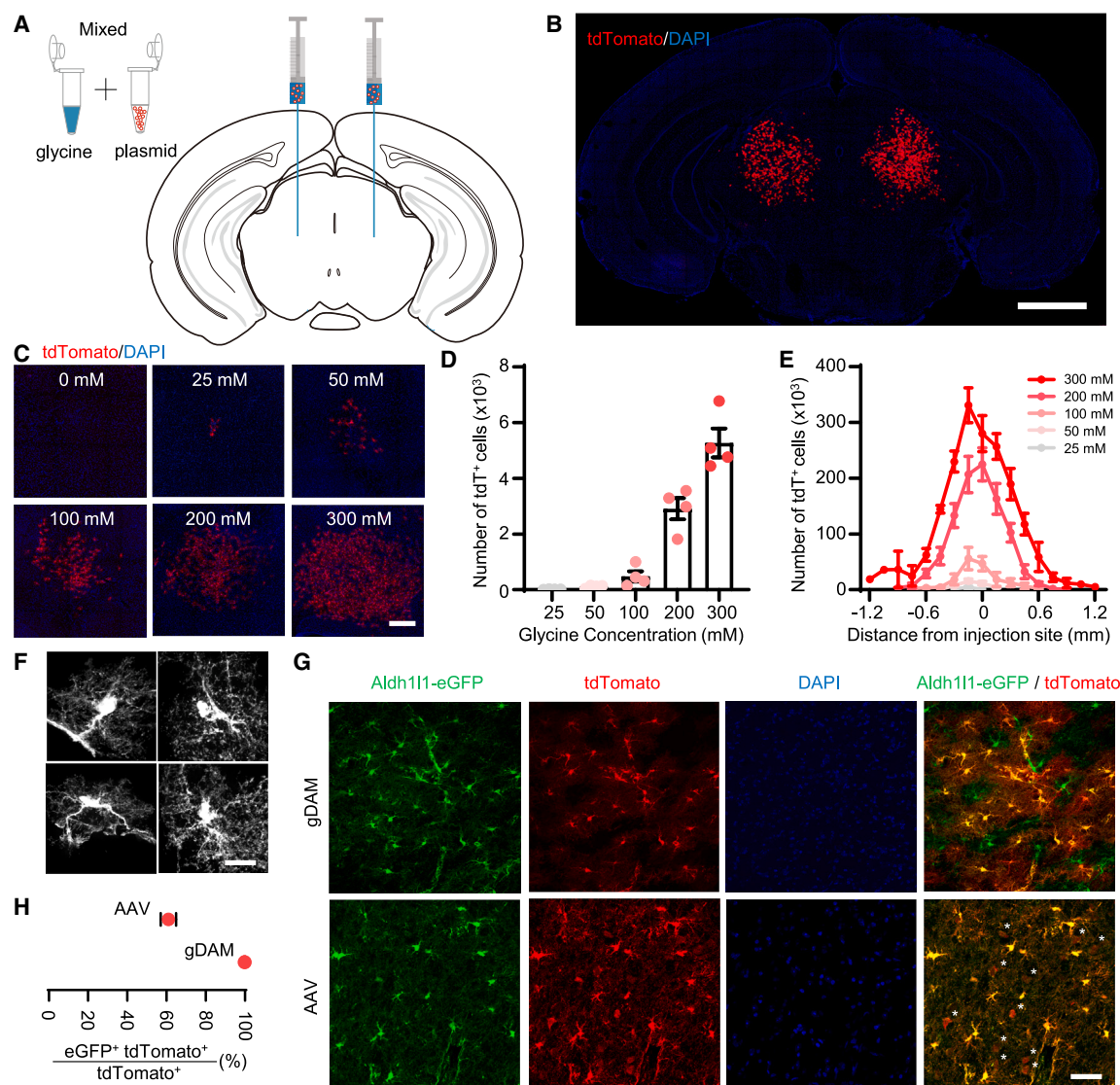


Figure 1. gDAM facilitates the rapid expression of naked DNA specifically in astrocytes

(A) Schematic illustration depicting the gDAM injection protocol, wherein a mixture of plasmid DNA and glycine solution was injected into the brain parenchyma of mice. (B) Representative images of fluorescently transfected cells using a mixture of plasmids encoding mScarlet (final concentration: 1 $\mu\text{g}/\mu\text{L}$) and glycine solution (final concentration: 300 mM). Scale bar, 1 mm. (C) Representative images of transfected tdTomato-labeled cells expressing Cre recombinase in Ai9 mice via gDAM with varying concentrations of glycine (0, 25, 50, 100, 200, and 300 mM). The hypotonicity of the gDAM delivery vehicles, which contained less than 300 mM glycine, was adjusted to normal physiological osmotic pressure by adding different concentrations of NaCl. Scale bar, 200 μm . (D) Total number of transfected tdTomato-labeled cells per injection course for each glycine concentration (25, 50, 100, 200, and 300 mM). (E) Number of transfected tdTomato-labeled cells in each 50- μm -thick brain slice for glycine concentrations (25, 50, 100, 200, and 300 mM). One out of every three consecutive slices was chosen for the quantification analysis. (F) Example images of transfected tdTomato-labeled cells illustrating the typical morphology of astrocytes. Scale bar, 20 μm . (G) Representative images of transfected tdTomato-labeled cells co-localized with astrocytes expressing eGFP in Aldh1l1-eGFP::Ai9 mice, as well as images illustrating the expression specificity of AAV vectors. Using AAV vectors, Cre recombinase was expressed in Aldh1l1-eGFP::Ai9 mice under the control of the astrocyte-specific promoter GfaABC1D. Asterisks indicate transfected tdTomato-labeled cells lacking eGFP expression. Scale bar, 50 μm . (H) Ratios of eGFP and tdTomato double-positive cells to tdTomato-positive cells for gDAM and AAV vectors, respectively (rAAV2/5-GfaABC1D-Cre, 3×10^{12} vg/mL).

neuronal (NeuN) (Figure S4B), oligodendrocyte precursor (NG2) (Figure S4C), and pre-myelinating and myelinating oligodendrocyte (CNPase) (Figure S4D) markers was performed, none of which were found to be positive in transfected tdTomato-labeled cells. However, nearly 20% of the gDAM-transfected cells were positive for

Olig2, albeit with faint staining levels (Figure S4E). Nonetheless, Olig2 staining in the crossbred mice of Aldh1l1-eGFP and Ai9 transgenic mice showed that all transfected cells with faint Olig2 staining were also eGFP positive (Figure S4F), indicating that these Olig2-positive cells were indeed astrocytes.

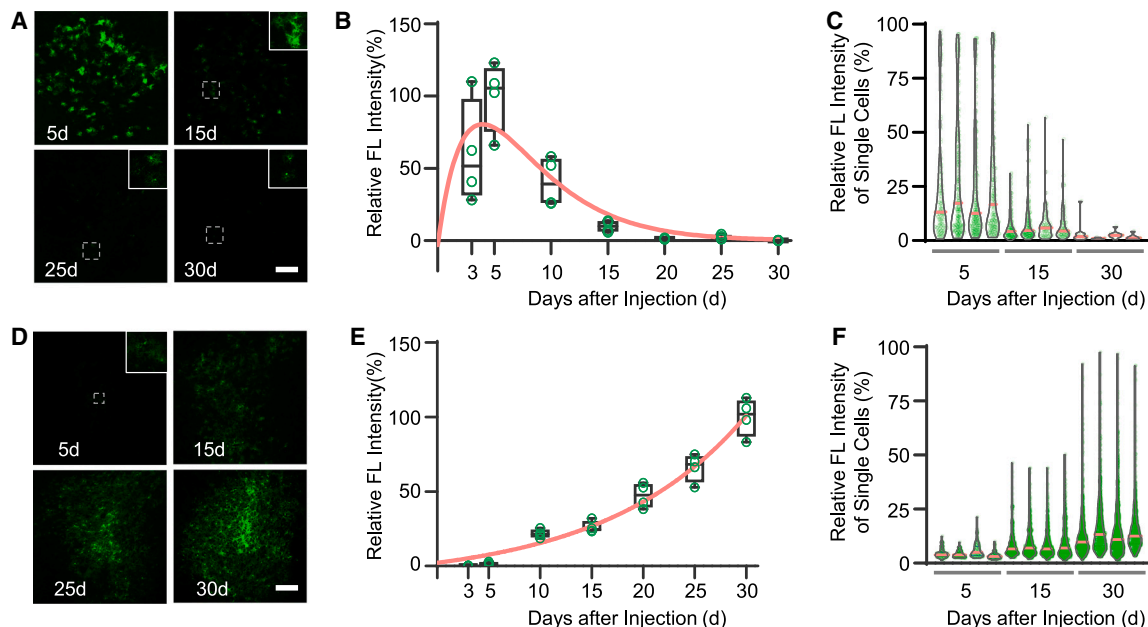


Figure 2. Dynamics of gDAM-mediated gene expression

(A and D) Representative images of eGFP-positive astrocytes after injection of the gDAM delivery vehicles (A) and AAV2/5 vectors (D) at different time points (5, 15, 25, and 30 days). The insets show enhanced brightness for visualizing the dim cells. Scale bar, 200 μ m. (B and E) Box plots illustrating total eGFP brightness at various time points (3, 5, 10, 15, 20, 25, and 30 days) after injection. The fitted curve is in red. Fluorescence intensity was normalized to the peak value at 5 days for gDAM (B) and 30 d for AAV2/5 (E). (C, F) Fluorescence intensity of individual single cells at 5, 15, and 30 days after injection of gDAM (C) and AAV virus (F) ($n = 4$ mice). The solid pink bar represents the median.

gDAM facilitates the rapid and transient expression of foreign genes in astrocytes

To elucidate the gene expression kinetic profiles of gDAM, we injected plasmid DNA expressing eGFP under the control of the CMV promoter and collected brain tissues at various intervals post-injection: days 3, 5, 10, 15, 20, 25, and 30 (Figure 2A). The fluorescence intensity of eGFP was quantified from 50- μ m-thick brain sections and subjected to exponential decay analysis.³⁴ The results showed that eGFP expression peaked at approximately day 4, followed by a decline to half-maximal fluorescence intensity by day 10.5, with the signal becoming negligible by day 25 (Figure 2B). Correspondingly, the fluorescence brightness of single astrocytes expressing eGFP aligned well with this temporal expression pattern on days 5, 15, and 30 after injection (Figure 2C). In parallel, AAV vectors expressing eGFP under the control of the GfaABC1D promoter were used. Very weak fluorescence intensity was observed on day 5 following the administration of AAV vectors (Figures 2D and 2F). However, the fluorescence intensity continued to increase and did not reach a plateau even by day 30 (Figure 2E). Collectively, these findings suggest that the delivery of DNA plasmid vectors by gDAM results in rapid and transient expression of foreign genes within astrocytes.

gDAM achieves long-term expression of foreign genes in astrocytes through piggyBac transposon-mediated genomic integration

Achieving long-term correction of genetic disorders requires the targeted delivery of therapeutic DNA to specific cells and tissues,

as well as ensuring stable expression at therapeutic doses. The piggyBac (PB) transposon system has emerged as a powerful method for achieving somatic integration. In this study, we engineered a donor vector containing eGFP within a transposable element, along with a helper vector encoding the transposase enzyme (Figure S6A). To test the ability of gDAM to deliver multiple plasmid vectors simultaneously, we first delivered two plasmid vectors expressing eGFP and tdTomato protein, respectively (Figures S5A–S5C) or three plasmid vectors expressing eGFP, TagBFP, and tdTomato protein (Figures S5D–S5F). We found that the co-expression efficiency of the two plasmids reached 73.5% (Figure S5C), while the co-expression efficiency of the three vectors reached 62.2% (Figure S5F). Next, we delivered the PB system via gDAM and measured fluorescence intensity at 5 and 25 days, observing a notable 2.6-fold increase in eGFP fluorescence intensity (Figure S6B). Continuous expression was observed even after six months, as indicated by the bright fluorescence of eGFP-labeled astrocytes. Notably, eGFP-positive cells were not observed after 6 months in the absence of transposase (Figure S6C). Furthermore, genomic DNA was extracted from brain tissues at the injection sites, and PCR amplification of the eGFP gene yielded a fragment of the desired size. In contrast, no PCR product was detected in the control group, which consisted of eGFP plasmid vectors alone without transposase expression (Figure S6D). These data indicate that PB delivered by gDAM successfully integrates foreign genes into the genome of astrocytes, thereby achieving long-term expression.

A diverse array of small metabolites facilitates the delivery of DNA plasmid vectors into astrocytes

Glycine is an important nutrient and 1 of the 20 basic amino acids that comprise proteins. However, it remains unclear whether other basic amino acids, similar to glycine, can also facilitate the delivery of naked DNA into brain cells. We substituted glycine in the gDAM with other L-type amino acids and found that most L-type amino acids were able to facilitate the delivery and expression of plasmid vectors to some extent. Notably, L-proline, L-serine, and L-histidine demonstrated comparable delivery efficiency to glycine. In contrast, no transfected tdTomato-positive cells were observed when using valine, phenylalanine, and methionine for DNA delivery (Figure 3A).

In addition to being a basic amino acid and a building block of proteins, glycine is also an important neurotransmitter and is required for multiple metabolic synthesis pathways. The diverse functionality of glycine prompted us to investigate the potential of non-protein amino acids in the delivery of naked DNA. Several D-amino acids have been identified in the brain and are known to have specific biological functions in mammals. Therefore, we assessed the delivery efficiency of D-type amino acids, and the results indicated that certain partial stereoisomers, particularly D-alanine, exhibited high delivery efficiency ($6,346 \pm 508$ cells) (Figures 3B and S7). Furthermore, a variety of metabolically synthesized non-protein amino acids, including gamma-aminobutyric acid (GABA), hypotaurine, taurine, sarcosine, β -alanine, L-hydroxyproline, L-ornithine, L-citrulline, and betaine, demonstrated significant delivery efficiency (Figures 3C and S8). Intriguingly, certain dipeptides and tripeptides also exhibited promising effects on delivery efficiency, such as Gly-Gly, Gly-Gly-Gly, and Leu-Gly. Notably, there was a significant difference in delivery efficiency between Leu-Gly and Gly-Leu as well as between Gly-His and His-Gly, despite only differing in the order of amino acids. Additionally, L-azetidine-2-carboxylic acid, a plant-derived L-proline analog, transfected a considerable number of tdTomato-positive cells ($2,244 \pm 77$ cells) (Figure S9). The differences in the efficiency of these amino acids in gene delivery may indicate that they have specific roles in the brain and astrocytes.

gDAM did not cause notable brain tissue damage

Considering that the concentration of these metabolites added to gDAM (300 mM) significantly exceeds typical physiological levels, there is a potential for damage. GFAP activation plays a critical role in astroglial activation under most pathological conditions including injury and neurodegeneration. After the injections, we monitored GFAP expression and found no discernible increase near the injection sites, suggesting negligible GFAP activation (Figure S10). Microglia can be activated during neuroinflammation and acute injuries. Examination of microglia using Iba1 immunostaining showed no significant change compared with PBS injections alone (Figure S11). Moreover, terminal deoxynucleotidyl transferase (TdT) dUTP nick-end labeling (TUNEL) staining revealed only isolated cells near the injection trajectory, likely attributable to the pas-

sage of the pipette (Figure S12). Ultimately, these experiments indicated no substantial tissue damage or inflammatory reactions attributable to gDAM.

gDAM was used to achieve CRISPR-Cas9-based genome editing

CRISPR-Cas9 is a highly effective gene editing tool that is widely used in scientific and clinical research. To enable CRISPR-Cas9-based genome editing, a large plasmid vector (approximately 10 kbp) was designed, containing three discrete open reading frames: CBh-Cas9, U6-single guide RNA (sgRNA) targeting the eYFP gene, and CMV-mCherry as the fluorescent reporter for Cas9 expression. A randomized sequence served as a non-targeting control. Another plasmid vector expressing Cre recombinase was used to induce eYFP expression in astrocytes in the midbrain of Ai3 mice (Figure 4A and Table S1). The targeted sgRNA bound to the Cas9 protein, resulting in the loss of function of eYFP, and leading to the exclusive expression of mCherry in astrocytes. In contrast, both mCherry and Cre recombinase-induced eYFP are co-expressed in astrocytes in the control group (Figure 4B). A quantification assay revealed that approximately 89.8% of transfected mCherry-labeled astrocytes in the sgYFP group were devoid of eYFP expression, compared with 11.3% in the control group (Figure 4C), indicating that CRISPR-Cas9 successfully induced mutations in the eYFP genes of astrocytes. Sanger sequencing was performed to identify sequence alterations, revealing that the delivery of CRISPR-Cas9 via gDAM induced frameshift indels at the targeted loci of eYFP genes in astrocytes (Figure 4D).

gDAM induced gliomas from mature astrocytes in immunocompetent mice

To investigate the potential of gDAM to induce gliomas in mouse brains by introducing multiple mutations into astrocytes, we simultaneously delivered four plasmids into the mouse midbrain. This included two plasmids encoding the Cas9 protein and sgRNAs targeting the tumor suppressor genes *Trp53*, *Pten*, and *Nf1* (Figure S13A), as well as two plasmids encoding the PB system for integrating the oncogene EGFRvIII into the chromosomal genome of astrocytes (Figure 4E). Tumors were detected in approximately 86.7% of mice (13 out of 15) and none survived beyond 9 months (Figures 4F and 4G). The progression of gliomas was monitored using magnetic resonance imaging (MRI), which revealed significant growth within a month (Figure 4H). Hematoxylin and eosin (H&E) staining further characterized the histological features of malignant glioblastoma multiforme, displaying hemorrhage, aneuploidy, and necrosis—hallmarks of malignant gliomas (Figure 4I). Glioma cells were isolated and cultured *in vitro*, with the primary glioma cells proliferating rapidly and multicellular tumorsphere-like structures observed at 14 days, indicating a degree of malignancy (Figure 4J). To further verify that the induced gliomas were derived from astrocytes, Sanger sequencing was performed on the cultured glioma cells, confirming genomic mutations in the tumor suppressor genes (*Trp53*, *Pten*, and *Nf1*) within the sgRNA-targeting region (Figure 4K and Table S2). Additionally, the mCherry

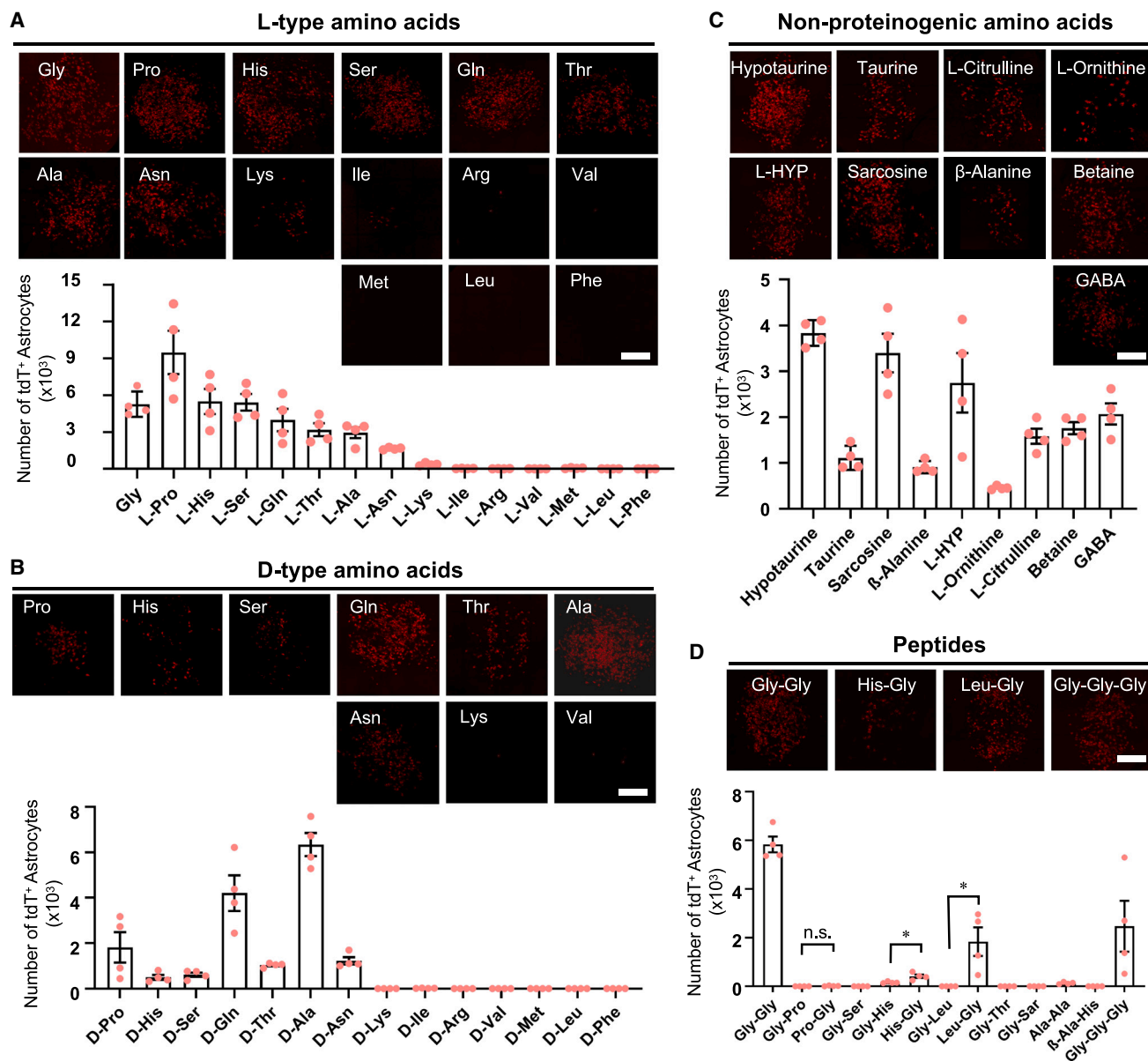
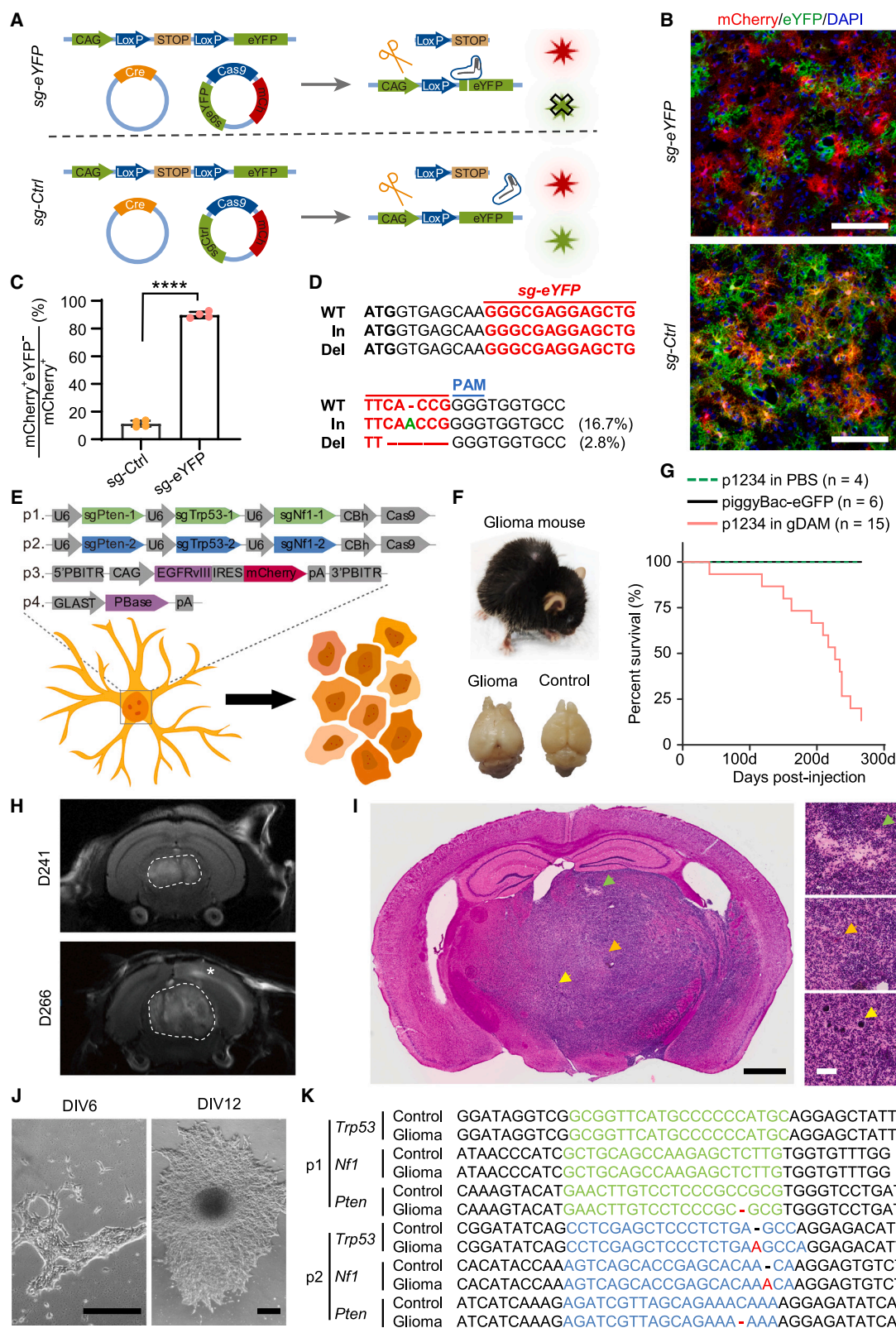


Figure 3. Other small metabolites also facilitate efficient delivery of naked DNA

(A) Representative images and the number of transfected tdTomato-labeled astrocytes using L-type proteinogenic amino acids as delivery vehicles. Amino acids in delivery vehicles have different concentrations: 300 mM (L-Pro, L-His, L-Ser, L-Gln, L-Thr, L-Ala, L-Asn, L-Lys, L-Arg, and L-Val), 260 mM (L-Ile), 220 mM (L-Met), 170 mM (L-Leu), and 100 mM (L-Phe). Other proteinogenic L-type amino acids have not been used as delivery vehicles because of their excitotoxicity or extremely low water solubility. Scale bar, 500 μ m. (B) Representative images and the number of transfected tdTomato-labeled astrocytes using D-type amino acids as delivery vehicles. The final concentrations of D-type amino acids were the same as that of the corresponding L-type. Scale bar, 500 μ m. (C) Representative images and the number of transfected tdTomato-labeled astrocytes using non-proteinogenic amino acids as delivery vehicles. The final concentrations of all nonproteinogenic amino acids were 300 mM. Scale bar, 500 μ m. (D) Representative images and the number of transfected tdTomato-labeled astrocytes using peptides as delivery vehicles. The final concentrations of all peptides used were 300 mM. Plasmid DNA (1 μ g/ μ L) encoding Cre recombinase was delivered to Ai9 reporter mice. Scale bar, 500 μ m. Gly-Pro vs. Pro-Gly, ^{n.s.}*p* = 0.056. Gly-His vs. His-Gly, **p* = 0.0206. Gly-Leu vs. Leu-Gly, **p* = 0.0141.

fluorescence signal was detected in the tumor region (Figures S13B and S13C), which was specifically designed to integrate into the genome via the PB transposon system along with the oncogene

EGFRvIII. These findings suggest that the induced gliomas in mice originated from astrocytes with genomic alterations driven by gDAM.



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gDAM-diacted over-expression of interferon- β and granulocyte-macrophage colony-stimulating factor in tumor-surrounding astrocytes prolongs the survival of glioma-bearing mice

Interferon- β (IFN- β) and granulocyte-macrophage colony-stimulating factor (GM-CSF) have been implicated in the immunotherapy of gliomas, raising the question of whether the overexpression of these cytokines in the astrocytes surrounding gliomas could enhance the survival of glioma-bearing mice. To verify the expression of foreign genes in the astrocytes of the periglioma, 4,000 eGFP⁺ 73C glioma cells, derived from an astrocyte (*BRAF*^{V600E}; *Pten*^{-/-}; *Trp53*^{-/-}), were transplanted into the brain parenchyma of C57BL/6J mice. Five days after implantation, gDAM delivery of a DNA plasmid vector expressing the fluorescent protein mScarlet was administrated (Figure 5A). Astrocytes at the periphery of the 73C glioma were fluorescently labeled, demonstrating that efficient astrocyte-specific uptake and expression of foreign genes are feasible in the periglioma region (Figure 5B).

To assess the therapeutic impact of IFN- β and GM-CSF, two plasmid vectors expressing these cytokines were designed and delivered to the glioma cells 5 days after implantation (Figure 5C). The results revealed that the simultaneous expression of both cytokines significantly prolonged the survival of glioma-bearing mice, improving the median survival to 26 days compared with 17 days in the control group (Figure 5D). The median survival was also improved in the IFN- β -expressing group (24 days) but not in the GM-CSF-expressing group (16 days) (Figure 5D), suggesting that the improved survival of glioma-bearing mice may be primarily mediated by IFN- β . To further investigate the therapeutic effects of AAV vectors that express both cytokines, two AAV vectors were design to express IFN- β and GM-CSF individually under the control of the GfaABC1D promoter (Figure S14). The co-expression of these two cytokines via viral vectors did not result in a significant improvement in the median survival time of glioma-bearing mice (20 days) (Figure 5D), and the relatively poor therapeutic effects may be attributed to low expression levels during the early expression stage of AAV. Next, the volumes of estab-

lished gliomas were measured by MRI at 9 and 14 days after implantation. At 14 days after implantation, the glioma volumes in the IFN- β + GM-CSF group were approximately half the size of those in the control group (control group, 18.3 ± 2.5 mm³; IFN- β + GM-CSF group, 8.7 ± 2.0 mm³) (Figure 5E).

To confirm whether T cells are responsible for inhibiting glioma growth, we performed immunofluorescence staining of CD4 and CD8. This revealed that an increased number of CD4- and CD8-positive T cells within the tumor area at 9 and 14 days after implantation (Figures 5F–5H and S15A). Flow cytometry analysis also indicated a significant increase in the proportion of CD4⁺ and CD8⁺ T cells relative to the total cell population at 15 days after implantation (Figure 5I). Taken together, these findings suggest that the simultaneous overexpression of IFN- β and GM-CSF in periglioma astrocytes may enhance the infiltration of CD4⁺ and CD8⁺ T cells, leading to inhibition of glioma growth.

DISCUSSION

Astrocytes are a major type of glial cell, and the genetic manipulation of astrocytes is essential for understanding their functions in physiology and pathology.³⁵ Here, we present an appealing approach for astrocyte-specific DNA delivery using a simple mixture of naked DNA and small metabolites. Until now, genetic manipulation targeting astrocytes has mainly relied on AAV vectors carrying astrocyte-specific promoters, commonly the GFAP promoter and the GfaABC1D promoter. Although significant efforts have been made to achieve more precise astrocyte-targeted expression, unwanted leaky expression remains an unavoidable issue. Inducible promoters and cell-state-specific promoters have also received increasing attention due to their important roles in biological research and gene therapy.^{36–39} However, developing conditional promoters that are both cell specific and cell state specific is undoubtedly more challenging. gDAM enables the precise astrocyte-specific DNA delivery without the limitations of cell-specific promoters, potentially facilitating

Figure 4. gDAM enables single plasmid-mediated genome editing and glioma model development

(A) Schematic diagram illustrating the strategy for CRISPR-Cas9 knock out the *eYFP* in Ai3 reporter mice using gDAM. The plasmid DNA consisted of two plasmid vectors: one (1,0006 bp) encoding the SpyCas9 protein under the CBh promoter, sgRNA under the U6 promoter, and reporter gene mCherry under the CMV promoter, and the other (6,422 bp) encoding Cre recombinase to activate the expression of eYFP in Ai3 mice. In control experiments, *sg-eYFP* was replaced with a random DNA sequence of the same length. These mixtures were individually injected into the midbrains of the Ai3 reporter mice. (B) Representative images showing the colocalization of transfected mCherry- and eYFP-labeled cells in CRISPR-Cas9 knockout and control mice. Brain sections were obtained at 7 days after injection. eYFP is shown as green in these cells, whereas yellow cells indicate the co-expression of both mCherry and eYFP. DAPI staining is shown in blue. Scale bar, 100 μ m. (C) Histogram of the proportion of eYFP-negative cells among all transfected mCherry-labeled cells ($n = 4$ mice). There were $89.9 \pm 2.2\%$ of transfected mCherry-labeled cells that were eYFP-negative in the *sg-eYFP* mice ($n = 4$), **** $p < 0.0001$. (D) Sanger sequencing analysis indicated that *sg-eYFP* successfully induced indel mutations at the target site of eYFP. sgRNA-targeting sequence is shown in red. Del, deletion; In, insertion mutation; PAM, protospacer-adjacent motif. (E) Strategy for developing a glioma model using gDAM. The four plasmids were mixed with gDAM delivery vehicles and delivered to the midbrains of C57BL/6J mice. (F) Images showing a mouse and brain affected by gliomas. (G) Kaplan-Meier curves for the overall survival of mice after plasmid injection. Plasmids p1234 in glycine solution were injected ($n = 15$ mice) and control group 1 was injected with plasmids p1234 in PBS ($n = 4$ mice), and control group 2 was injected with PB-eGFP (p4 + p5: PB-CAG-eGFP) in glycine solution ($n = 6$). Log rank test, **** $p < 0.0001$, p1234 in gDAM vs. PB-eGFP or p1234 in PBS. (H) T2-weighted MRI of the same mouse in (F) with gliomas in the brain on days 241 and 266. The intensity of T2 signal in the midbrain increased dramatically within 26 days. Asterisk: T2-weighted MRI showing a hyperintense lesion in the corpus callosum with rapid glioma progression. (I) H&E staining of gliomas brain section. The brain was found to be malformed. Typical pathological features of malignant glioma, including necrotic region (green arrow), intratumoral hemorrhage (orange arrow), and giant aneuploid cells with pleomorphic nuclei (yellow arrow) are shown. Scale bars, 1 mm and 50 μ m (on the magnified images). (J) Bright-field images of primary glioma cells from glioma mice, as shown in (J) at two different time points. DIV, days *in vitro*. Scale bar, 300 μ m. (K) Mutation validation of the cultured glioma cells in (J) by sanger sequencing showing a frameshift mutation that occurred in all three tumor suppressor genes (*Trp53*, *Nf1*, and *Pten*).

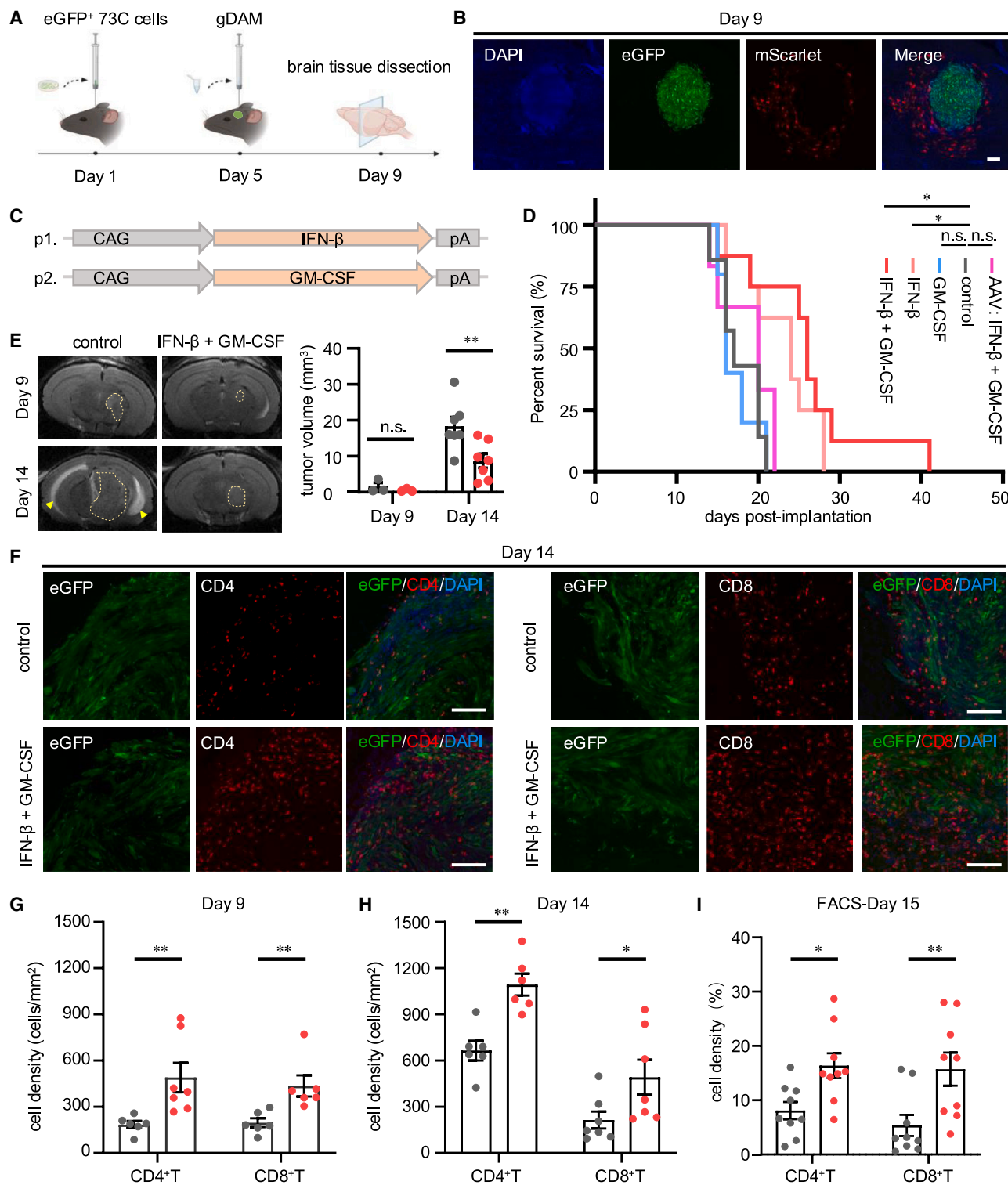


Figure 5. Over-expression of IFN-β and GM-CSF in the peritumoral astrocytes prolonged the overall survival of glioma-bearing mice

(A) Schematic schedule for implantation of glioma cells and overexpression of therapeutic proteins by gDAM. Gliomas were generated by intracranial implantation of 4,000 eGFP⁺ 73C glioma cells in C57BL/6J mice. Astrocyte-targeting delivery of plasmid vectors by gDAM was performed at day 5 after implantation of glioma cells. (B)

(legend continued on next page)

conditional gene expression in astrocytes *in vivo*. Crucially, by avoiding the use of viral agents and synthetic nanocarriers, gDAM could mitigate associated risks and drawbacks, such as immunogenicity, cytotoxicity, and complexity, thereby offering a streamlined and biocompatible alternative for the delivery of therapeutic genes.

Initially, glycine was identified as an efficient vehicle for delivering naked DNA to astrocytes, with delivery efficiency dependent on its concentration. A few astrocytes were sparsely transfected at glycine concentrations as low as 25 mM. Glycine is 1 of the 20 basic amino acids that make up proteins; these proteinogenic amino acids have identical backbones but variable side chain R-groups. Among these structurally similar amino acids, glycine, serine, proline, and histidine demonstrated comparable efficiency in terms of DNA delivery. Additionally, non-essential amino acids seem to be more efficient than essential amino acids (Figure S7). Certain amino acids, such as glutamate, cysteine, and aspartate, were excluded from the experiments due to excitotoxicity,^{40–43} while tryptophan and tyrosine were not included because of their low water solubility. A comparison of L-type and D-type amino acids in terms of delivery efficiency revealed that most L-type amino acids had better efficiency than D-type amino acids, except for D-Ala, which surprisingly showed higher efficiency than L-Ala. Some D-amino acids have been discovered in mammalian brains and are emerging as a new and significant class of signaling molecules.⁴⁴ Our results suggest that D-type amino acids may play potential roles in astrocytes. We further examined other intracellular non-proteinogenic amino acids in the brain. GABA is the primary inhibitory neurotransmitter, and taurine, hypotaurine, sarcosine (N-methylglycine), betaine (trimethylglycine), and β -alanine have also been shown to have neuroprotective effects in the CNS.^{45–49} In this study, these non-proteinogenic amino acids also demonstrated high efficiency for the delivery of naked DNA. Notably, we found that some oligopeptides such as Gly-Gly, Gly-Gly-Gly, and Gly-Leu, exhibited similar effects. Significant differences in delivery efficiencies were observed between Gly-Leu and Leu-Gly, while Gly-Pro and Pro-Gly were found to be completely ineffective in delivering naked DNA. This suggests that these peptides themselves played important roles in the delivery of naked DNA, rather than the amino acids resulting from the potential degradation of these peptides. Collectively, diverse small molecules have demonstrated astrocyte-

targeted delivery of naked DNA, and it is reasonable to speculate that other small molecules with similar functions likely exist. However, the mechanism behind the specific targeting of astrocytes and the variable delivery efficiency of small metabolites remains unclear, warranting further research.

Gene therapy is a promising therapeutic modality for correcting many genetic disorders.⁵⁰ The potential off-target outcomes of gene editing raise significant concerns in clinical applications,⁵¹ and the long-term expression of genetic manipulation tools undoubtedly increases off-target risks. Using gDAM delivery, we measured the expression level of eGFP driven by the ubiquitous CMV promoter at different time points. The results showed that eGFP expression peaked on the fourth day, with no expression observed on the 25th day. Sustained, even permanent, therapeutic effects of gene therapy are particularly appealing for treating neurological disorders in certain cases. Integrating foreign DNA into the genome of host cells is an important method of gene therapy for achieving stable and long-term expression, and significant efforts have been made to develop and improve genomic integration techniques in recent years.^{52,53} PB is a highly efficient genomic integration tool that delivers large transgenes through a two-vector transposon system, which consists of a transposase vector and a donor vector containing the gene of interest flanked by inverted terminal repeats.⁵⁴ In this study, PB was applied to successfully integrate eGFP into the astrocyte genome to achieve long-term expression. Additionally, a notable obstacle of AAV vectors is their small packaging capacity (approximately 4.7 kb in most cases).¹² In contrast with viral vectors, nonviral delivery vehicles are generally considered to have a higher DNA loading capacity. Using gDAM delivery, we constructed easy and efficient all-in-one CRISPR-Cas9 vectors to knock out the eYFP gene in the astrocytes of Ai3 transgenic mice. Furthermore, by combining CRISPR-Cas9 with the PB system, we accomplished mutations in tumor suppressor genes (*Nf1*, *Pten*, *Trp53*) and overexpression of the proto-oncogene (EGFRvIII) in immunocompetent mice, successfully inducing astrocyte-derived gliomas. This glioma model has two outstanding traits: (1) the cellular origin of gliomas is clear, which may contribute to the study of gliomagenesis and the early progression of astrocyte-derived gliomas, and (2) owing to the various driver genes for tumorigenesis, flexible mutation combinations allow for

Representative images of the delivery of plasmid vectors overexpressing mScarlet fluorescent proteins using gDAM at day 5 after implantation of glioma cells. Astrocytes (red) at periphery of the 73C glioma (green) were transfected successfully. The mice were sacrificed 9 d after glioma cells implantation. Scale bar, 200 μ m. (C) Schematic diagram of the therapeutic plasmid vectors overexpressing IFN- β and GM-CSF. (D) Kaplan-Meier survival curve of mice bearing eGFP⁺ 73C glioma in the control group (expressing mScarlet protein, $n = 7$), IFN- β group ($n = 5$), GM-CSF group ($n = 8$), IFN- β + GM-CSF group (co-expressing both IFN- β and GM-CSF, $n = 8$), and AAV: IFN- β + GM-CSF group ($n = 6$). The p value was calculated by using log rank Mantel-Cox test. Control vs. GM-CSF, $n.s.p = 0.7415$. Control vs. IFN- β , $*p = 0.0292$. Control vs. IFN- β + GM-CSF, $*p = 0.0202$. Control vs. AAV: IFN- β + GM-CSF, $n.s.p = 0.2598$. (E) T2-weighted MRI of the brain of the control and IFN- β + GM-CSF groups at 9 and 14 d after implantation of glioma cells, and assessment of the tumor volume based on hyperintense MRI signals. The dashed lines (yellow) delineate the tumor areas. Arrow heads show the lesions caused by rapid tumor progression. $n.s.p = 0.4$ (day 9) and $**p = 0.0076$ (day 14). (F) Shown are representative immunofluorescence images of CD4 (left) and CD8 (right) in the control group and IFN- β + GM-CSF group at 14 days post-implantation of eGFP⁺ 73C glioma cells (green). Scale bar, 100 μ m. (G) Quantitative analysis of CD4⁺ and CD8⁺ T cell infiltration into transplanted gliomas at 9 days post-implantation in the control and IFN- β + GM-CSF groups. $**p = 0.0012$ (CD4⁺ T) and $**p = 0.0022$ (CD8⁺ T). (H) Quantitative analysis of CD4⁺ and CD8⁺ T cells infiltration into the transplanted gliomas at 14 days post-implantation in the control group and IFN- β + GM-CSF group. $**p = 0.0043$ (CD4⁺ T) and $*p = 0.0262$ (CD8⁺ T). (I) Flow cytometry analysis of CD4⁺ and CD8⁺ cell proportions relative to tumor cells in eGFP⁺ 73C glioma at 15 days after implantation of glioma cells. Results are expressed as the mean $\pm$ SEM of the percentage of CD4⁺ or CD8⁺ T cells/tumor cells. $*p = 0.0176$ (CD4⁺ T) and $**p = 0.0040$ (CD8⁺ T).

exploration of the effects of different mutations on tumorigenesis in a relatively easy and cost-effective manner.

Immunotherapies are emerging as promising avenues that may offer new hope for the treatment of gliomas.^{55,56} Tumor-associated astrocytes play an important role in modulating the immune microenvironment of gliomas.^{6,7} It has been described that the function of IFN- β and GM-CSF in inhibiting tumor growth requires T cells.⁵⁷ In this study, periglioma astrocytes are used as a source of therapeutic proteins. Rapid and robust expression of IFN- β and GM-CSF in periglioma astrocytes through gDAM improved the infiltration of CD4⁺ and CD8⁺ T cells, thereby slowing glioma progression. This strategy could be harnessed as an alternative and adjunctive approach to the therapeutic treatment of gliomas.

In summary, gDAM is a novel approach that uses small molecule metabolites to effectively deliver naked DNA specifically into astrocytes. By avoiding the use of viral vectors and the incorporation of nanomaterials, gDAM circumvents the limitations and potential complications associated with these materials. The innovative strategy behind gDAM is expected to inspire new avenues in the search for gene delivery in other tissues, such as skeletal muscle and skin, which are important targets for vaccines and gene therapy.

MATERIALS AND METHODS

Mice

Mice were group housed, with up to five mice per cage, under a 12-h light/dark cycle (lights on from 7:00 AM to 7:00 PM). They had ad libitum access to water and food in an environment maintained at 22°C–25°C and 50% relative humidity. The adult C57BL/6J, Aldh1l1-eGFP (Jax Stock No. 012704), Ai3 (Jax Stock No. 007903), and Ai9 (Jax Stock No. 007909) mice were used in the experiments. All experimental procedures were approved by the Institutional Animal Care and Use Committee at the Chinese Institute for Brain Research.

Stereotaxic injection of plasmid and AAV vectors

Mice were anesthetized with pentobarbital (100 mg/kg, intraperitoneally) and positioned in a stereotactic apparatus with a heating pad maintained at 37°C. Their eyes were protected with chlortetracycline hydrochloride eye ointment. The skull surface was cleaned with 3% hydrogen peroxide to remove any fascia. The bregma and lambda points were used to adjust the mouse's head to a horizontal position. Using a microsyringe pump, 1 μ L of the gDAM mixture was slowly injected at a rate of 1 nL/s into the targeted brain areas. Each mouse was injected with a total of 1 μ L of AAV vectors expressing IFN- β (1.5×10^{12} vg/mL) and GM-CSF (8.3×10^{11} vg/mL), with 500 nL for each vector. The glass pipettes were held in place for at least 5 min after the injection before being withdrawn.

Stereotaxic injection of eGFP⁺ 73C glioma cells

Each C57BL/6J mouse received a stereotactic injection of 4,000 eGFP⁺ 73C tumor cells into the midbrain at a rate of 5 nL/s. The tumor cells treated with acridine orange/propidium iodide staining were accurately counted using a cell counter to ensure consistency among the

model mice. After the counting process, the cells were promptly placed on ice and injected within a 2-h window to maintain their viability. The injections were performed at the following coordinates: AP 3.2 mm, ML 1.2 mm, and DV 3.6 mm. On the fifth day after tumor cell transplantation, therapeutic plasmids mixed with a 300 mM proline solution (1 μ g) were injected at the same coordinates.

Histology and imaging

At the specified times following gDAM delivery, mice were perfused with PBS followed by 4% paraformaldehyde in PBS. The brains were removed, post-fixed overnight in 4% paraformaldehyde in PBS, and dehydrated in 30% sucrose for 24 h. The brains were sectioned in the coronal orientation at a thickness of 50 μ m for cell counting and immunostaining. Fluorescence images were acquired using a Leica SP8 confocal microscope with a 20 $\times$ objective, or a Nikon A1R confocal microscope equipped with a 20 $\times$ oil objective (CFI Plan Fluor NA 0.75 $\times$) and 40 $\times$ oil objective (CFI Plan Fluor NA 0.95). Images were analyzed using ImageJ software.

To quantify the fluorescence intensity corresponding to the activation of microglia (Iba1 staining) and astrocytes (GFAP staining) around the injection sites, we obtained confocal images from three consecutive slices of each mouse brain. We used a consistent set of imaging parameters—such as laser power, pinhole, PMT gain, offset value, and filter settings—to enable comparative measurements across images. The brightest sample from all mice was used to define the optimal confocal settings. We then applied these settings to the acquisition of the remaining images to prevent overexposure and ensure the accuracy of fluorescence intensity measurements. To quantify the density of Iba1-positive cells around the injection sites, we performed a three-dimensional reconstruction of three areas of the slice, measuring a volume of $318 \times 318 \times 50$ μ m around the injection sites (using a 40 $\times$ oil objective) for four injection sites from two animals. One of every three consecutive brain slices was selected for cell counting. All tdTomato and DAPI double-positive cells were counted, and the total number of transfected cells per brain was calculated by multiplying the total number of cells counted within the sections by three.

Quantification of fluorescent protein expression

All images were obtained using the same set of imaging parameters, including laser power, PMT gain, offset value, and filter settings. These parameters were predetermined using the brightest brain slice among all mice to prevent potential saturation. Protein expression was quantified as the average intensity in arbitrary units and as the percentage area of the sample exhibiting immunofluorescence that was 3 standard deviations above the background mean.

Evaluation of multi-plasmid co-expression

Mice were sacrificed 5 days after gDAM administration of two or three fluorescent protein plasmids, *p*-eGFP-c1, *p*-TagBFP, and *p*-tdTomato-N1, each of which was adjusted to a working concentration of 1 μ g/ μ L. Mouse brains were prepared as described above and sectioned at a thickness of 50 μ m on a freezing microtome. Confocal fluorescence images were acquired by using a Nikon A1R laser scanning confocal

microscope. The fluorescence intensity of each astrocyte was decomposed into RGB channels and then represented based on the average intensity above the threshold of the corresponding image stack from a region of interest within each astrocyte soma. The threshold was defined as 3 standard deviations above the background mean of each channel. If the value of a channel was less than this threshold, the corresponding fluorescent protein was not expressed in the cell.

Immunohistological staining

Serial coronal sections, 16 μm or 40 μm thick, were cut to encompass the entire region surrounding the injection sites. Immunostaining was conducted on free-floating sections, which were first incubated in a blocking solution consisting of 5% (v/v) BSA in PBS with 0.3% Triton X-100 for at least 1 h at room temperature. Subsequently, the sections were incubated in the same blocking solution for at least 1 h at room temperature, followed by incubation with primary antibodies at 4°C on a shaking platform. Primary antibodies included mouse anti-NeuN antibody (Millipore; 1:300, incubated for 48 h), rabbit anti-GFAP antibody (Sigma-Aldrich; 1:300, incubated for 16 h), chicken anti-Olig2 antibody (Abcam; 1:300, incubated for 16 h), mouse anti-CNPase antibody (Abcam; 1:500, incubated for 48 h), rabbit anti-Iba1 antibody (Wako Chemical; 1:1000, incubated for 16 h), rabbit anti-NG2 antibody (Millipore; 1:500, incubated for 48 h), rabbit anti-Ki67 antibody (Abcam; 0.5 $\mu\text{g}/\text{mL}$, incubated for 16 h), rabbit anti-CD8 antibody (Abcam; 1:500, incubated for 48 h), and rat anti-CD4 antibody redFluor 710 conjugate (Cell Signaling Technologies; 1:400, incubated for 48 h). After three washes with $1 \times$ PBS, sections were incubated with secondary antibodies for 2 h at room temperature. Secondary antibodies included goat anti-rabbit IgG conjugated to Alexa Fluor 488, goat anti-mouse IgG conjugated to Alexa Fluor 488, goat anti-chicken IgG conjugated to Alexa Fluor 488, goat anti-mouse IgG conjugated to Alexa Fluor 546, goat anti-chicken IgG conjugated to Alexa Fluor 568, goat anti-chicken IgG conjugated to Alexa Fluor 647, goat anti-rabbit IgG conjugated to Alexa Fluor 546, goat anti-rabbit IgG conjugated to Alexa Fluor 647, and goat anti-rat IgG conjugated to Alexa Fluor 546. Finally, brain sections were stained with DAPI solution at 1–5 $\mu\text{g}/\text{mL}$ and then rinsed with $1 \times$ PBS. Sections were mounted on slides, coverslipped with Vectashield mounting medium (Vector Laboratories), and sealed with nail polish.

TUNEL and H&E staining

All sections were fixed with 4% paraformaldehyde for 10 min at room temperature. The sections were processed with 0.1% Triton X-100 for 5 min, followed by a blocking solution for 1 h at room temperature. The slides were then transferred to TUNEL (Biotium, 30074) working solution for 1 h at 37°C and then rinsed. Nuclei were counterstained with DAPI and coverslips were sealed in place with nail polish. The H&E staining was performed using a Hematoxylin-Eosin Stain Kit (Solarbio, G1120).

Genomic DNA extraction and sequencing analysis of brain tissue

Genomic DNA from the brain tissue around the injection site was extracted by using a DNeasy Blood & Tissue Kit (QIAGEN, 69504). The

genomic region flanking the CRISPR target site was PCR-amplified using Herculase II high-fidelity polymerase (Agilent Technologies). PCR products were gel-purified and cloned using a Zero Blunt TOPO PCR Cloning Kit (Life Technologies, K2875-J10). Mutations were identified by Sanger sequencing.

The culture of primary glioma cells

We obtained 400- μm brain slices using a vibrotome (Leica VT1200). Tissues from the glioma region were isolated and cut into small pieces with sterile scissors in Hibernation buffer containing 1 mg/mL pronase and 2.5 $\mu\text{g}/\text{mL}$ DNase. The tumor pieces were incubated at 37°C for 30 min and pipetted gently 5–10 times, followed by termination of digestion of the glioma tissues with 10% FBS (Gibco, 10099141). Subsequently, the tumor tissue homogenate was centrifuged in a 10% density gradient medium (Gigma, D1556) at $600 \times g$ for 10 min. After discarding the supernatant, cells were resuspended and cultured in glioma medium containing neurobasal; glutamine, penicillin/streptomycin, sodium pyruvate, gentamycin; Trace Elements B, D-Biotin, SATO (Transferrin; BSA; Putrescine; Progesterone; sodium selenite), insulin, B27 minus VA, epidermal growth factor/fibroblast growth factor, and platelet-derived growth factor-AA.

The culture of eGFP⁺ 73C glioma cells

The eGFP⁺ 73C glioma cells, derived from primary mouse astrocytes and harboring mutations including *Trp53*^{-/-}, *Pten*^{-/-}, and *Braf*^{V600E}, were cultured in DMEM supplemented with 10% fetal bovine serum.

MRI evaluation

The MRI experiments were conducted using a Bruker 7.0 T MRI scanner (Bruker Pharmascan 70/60) with a slice thickness of 0.5 mm. The scanner was equipped with a 23-mm surface coil and a 12-cm diameter self-shielded gradient system for signal acquisition. The user interface was operated using Paravision 5.1 software (Bruker BioSpin) on a Linux PC running Topspin 2.0. Anesthesia was induced using a mixture of 5% isoflurane and 95% O₂ in a Plexiglass chamber, followed by maintenance with a mixture of 2% isoflurane and 98% O₂ at the magnetic field center. The respiratory rate of mice was monitored using an animal physiological monitoring system. Glioma volume was quantified from T2-weighted images using the Insight Segmentation and Registration Toolkit Simple Neuron Analysis of Pathology software.

Flow cytometry

Tumor-bearing mice were deeply anesthetized for euthanasia and immediately perfused with PBS. Subsequently, eGFP-positive tumor tissues were dissected under a Leica fluorescent stereomicroscope, finely minced using ophthalmic scissors, and digested in a solution containing papain (20–30 U/mL, Worthington Biochemical, LS003127) and DNAase (20 U/mL, Sigma, 10104159001) at 37°C for 30 min to obtain a single-cell suspension. Each sample was resuspended in 1 mL PBS, and Fc block buffer (Purified Rat Anti-Mouse CD16/CD32; BD, 553142) was added at a concentration of 1 μg per 1 million cells, followed by a 5-min incubation at 4°C. The samples

were incubated with the following antibody combination without washing off the Fc block buffer for 30 min at room temperature: CD45/APC-Cyannine7 (1.25 $\mu\text{L}/10^6$ cells; Biolegend, 103116), CD3/Brilliant510 (2.5 $\mu\text{L}/10^6$ cells; Biolegend, 100353), CD4/PE (1.25 $\mu\text{L}/10^6$ cells; Biolegend, 100407), and CD8/APC (1.25 $\mu\text{L}/10^6$ cells; Biolegend, 140410). After staining, the samples were washed twice with PBS and stained with DAPI before analysis on the FACSaria Fusion (BD) using FACSDIVA and FlowJo v10 software. The gating strategy used during the analysis is shown in Figure S15B.

Data analysis

All statistical analyses were performed using GraphPad Prism software or MATLAB R2016b (MathWorks). The data were considered to be statistically significant at $p < 0.05$. Asterisks denote statistical significance: $*p < 0.05$, $**p < 0.01$, $***p < 0.001$, $****p < 0.0001$. Unless otherwise indicated, values are reported as the mean $\pm$ SEM.

DATA AND CODE AVAILABILITY

The authors confirm that the data supporting the findings of this study are available within the article and its [supplemental information](#).

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AUTHOR CONTRIBUTIONS

W.S., J.H., and H.Z. conceived the study. H.Z. and J.D. conducted most of the experiments. D.L. and L.W. contributed to DNA purification. X.H. and M.Y. helped conduct the immunostaining and the culture of primary glioma cells. J.L. contributed to a critical reagent. H.Z., J.D., and W.S. analyzed the data. W.S. and H.Z. wrote the manuscript with feedback from the other authors.

DECLARATION OF INTERESTS

The Chinese Institute for Brain Research, Beijing, has filed patent applications related to this work with W.S., J.H., and H.Z. listed as inventors.

SUPPLEMENTAL INFORMATION

Supplemental information can be found online at <https://doi.org/10.1016/j.ymthe.2025.01.006>.

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